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Effect of NPK fertilizer, biofertilizer containing N fixer and P solubilizer, and green manure of *C. juncea* on nutrients uptake and growth of sugarcane

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Abstract. A field experiment was conducted at Karangploso Research Station to investigate the effect of NPK fertilizer added with biofertilizer containing of N fixing and P solubilizing bacteria plus green manure of *C. juncea* on nutrient uptake and growth of sugarcane (*Saccharum officinarum* L.). The experiment had 10 treatments: (1) 100% of recommended fertilizer (RF), (2) 100% RF + biofertilizer, (3) 75% RF + biofertilizer, (4) 50% RF + biofertilizer, (5) 100% RF + *C. juncea*, (6) 75% RF + *C. juncea*, (7) 50% RF + *C. juncea*, (8) 100% RF + biofertilizer + *C. juncea*, (9) 75% RF + biofertilizer + *C. juncea*, (10) 50% RF + biofertilizer + *C. juncea*. The ten treatments were arranged in Randomized Block Design with three replications. Results showed that the nutrients uptake attributes (N, P and K) and growth of sugarcane was affected by biofertilizer and *C. juncea*. The best treatment of the results was combination of biofertilizer and *C. juncea* with 100% rate of recommended fertilizer of NPK gave the highest of N uptake and length of stalk.

Keywords: Biofertilizer, green manure, inorganic fertilizer, sugarcane, nutrients

1. Introduction

Sugarcane is categorized as high fertilizer consumption crop that requires a lot amount of nutrients to have good growth and high yield [3]. The nutrients of 135 kg N ha^{-1}, 32 kg P ha^{-1} and 98 kg K ha^{-1} were required to produce 80 tons ha^{-1} of sugarcane in the first cycle [14]. Mostly the source of nutrients added to sugarcane cropping is inorganic fertilizers and continuous addition of inorganic fertilizers may lead to environmental problems due to the leaching of soluble nutrients into deeper soil layers [20], soil degradation, soil acidity and environmental pollution [17].

The integrated nutrient management practice is an important factor for the sustainable and cost-effective management of soil fertility in sugarcane cultivation. Mixed addition of inorganic with organic materials (including biological fertilizers) may have result in rising soil fertility, growth and yield of sugarcane.

Some previous research reported the positive effect of biofertilizer and organic materials either on soil fertility or on plant growth and yield. Previous research found that addition of biofertilizer containing *Acidithiobacillus* bacteria increased nutrient uptake of sugarcane and raised the availability of nutrients in the soil, especially total N, nitrate-N, ammonium-N, and available P and K [14]. The effects of biofertilizer were enhanced by the combination addition with organic fertilizer in the form of

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sugarcane filter cake. The effectiveness of biofertilizer mixed with organic matter has been also reported in research using annual crops, such as sugarcane [21; 14], rice [1; 9; 4]; and Zea mays [8; 12]. However, most of the research was conducted by using single bacteria which has role to increase the availability of a certain nutrient. Mixed inoculant of N2-Fixing, P- and K-Solubilizing bacteria was able to improve available soil N, P, and K and kiwifruit growth [19]. So that there is a chance to use biofertilizer containing mixed inoculant of N_2 fixing and P-solubilizing bacteria on nutrients uptake and growth of sugarcane.

The effectiveness of the biofertilizer on soil fertility and plant growth may be raised when the application of biofertilizer is combined together with green manure, such as *C. juncea*. Some previous studies reported the positive effect of *C juncea* on enhancing soil fertility, growth and yield of sugarcane ([; 10; 18]. The objective of the research was to investigate the effect of NPK fertilizer added with mixed inoculant of N fixer and P solubilizer bacteria plus green manure *C juncea* on nutrients uptake and growth of sugarcane.

2. Materials and methods

The field research was conducted at Karangploso Research Station of Indonesian Sweetener and Fiber Crops Research Institute in 2016. Biofertilizer containing N fixing and P solubilizing bacteria, *C. juncea*, and inorganic fertilizer were used as treatments. The field soil used in the research is categorized as low in soil nutrients contents as indicated by very low of organic-C, low in total N, P and K contents (Table 1). The texture of the soil is silty clay with sand, silt and clay contents of 6%, 45% and 49% respectively.

Table 1. Nutrients contents of soil experiment

Soil Nutrients	Content	Category [6]
Organic-C (%)	1.56	Very low
Total N (%)	0.14	Low
C/N	11	Low
P_2O_5 Bray (mg.kg^{-1})	3.05	Low
K NH_4OAC1N pH:7 (me/100g)	0.12	Low

The treatments consisted of 10 combination of biofertilizer, inorganic fertilizer, and *C juncea*, and their composition applications were as follow:

1. 100% recommended fertilizer of sugarcane (RF: NPK 600 kg ha^{-1}+ Ammonium Sulphate 500 kg ha^{-1})
2. 100% RF + biofertilizer (BF)
3. 75% RF + BF
4. 50% RF + BF
5. 100% RF + *C juncea*
6. 75% RF + *C juncea*
7. 50% RF + *C juncea*
8. 100% RF + BF + *C juncea*
9. 75% RF + BF + *C juncea*
10. 50% RF + BF + *C juncea*

Those treatments were arranged using Randomized Block Design with three replications. Sugarcane var BL was planted in plots size 7.5 m x 7 m with row spaces of 125 cm. Bagal seeds with three buds were planted at rows. Recommended fertilizer (RF) was applied twice, firstly 600 kg NPK fertilizer per hectare (15 N : 15 P_2O_5 : 15 K_2O) was added at one month after planting and secondly 500 kg Ammonium Sulphate (21% N) per hectare was added at 3 months after planting. *C juncea* was planted at 20 days before planting of sugarcane. The seeds of *C juncea* were placed along the rows of sugarcane and after 40 days, *C juncea* was cut and chopped and then finally they were mixed and buried into the

soil at weeding. The microbial inoculants were collection of Indonesian Sweetener and Fiber Crops Research Institute which were isolated from rhizosphere of sugarcane soil. Biofertilizer was applied at 2 months after planting of sugarcane at the rate of 100 kg/ha with bacterial population of 11.3 x 10 8 cfu gram^{-1}.

Observation was made on nutrients uptake and sugarcane growth parameters. Nutrients uptake was determined after sugarcane was harvested by analyzing the content of N, P and K in dried biomass. Growth of sugarcane parameters that were measured included length of stalk, plant population, diameter of stalk and number of internodes. Collected data were analyzed using Anova and followed by analysis of treatments comparison by using Least Significant Different method at 5% level.

3. Results

3.1. N, P, K Nutrients Uptakes

Significant effect of the treatments ($p < 0.001$) was occurred in N, P, and K uptake by sugarcane (Figure 1). In general, addition of NPK fertilizer and biofertilizer with or without *C juncea* raised the uptake of N and P nutrients compared to control (except treatment 3, 6 and 7 for N uptake and treatment 4 and 6 for P uptake). For K uptake, only treatments 4 and 8 were significantly higher than control. The highest absorption of N (41.16 g plant^{-1}) by sugarcane was found at treatment 8 (100% RF + BF + *C juncea*), and if compared to the control (100% RF), it can increase N uptake of about 62.56%.

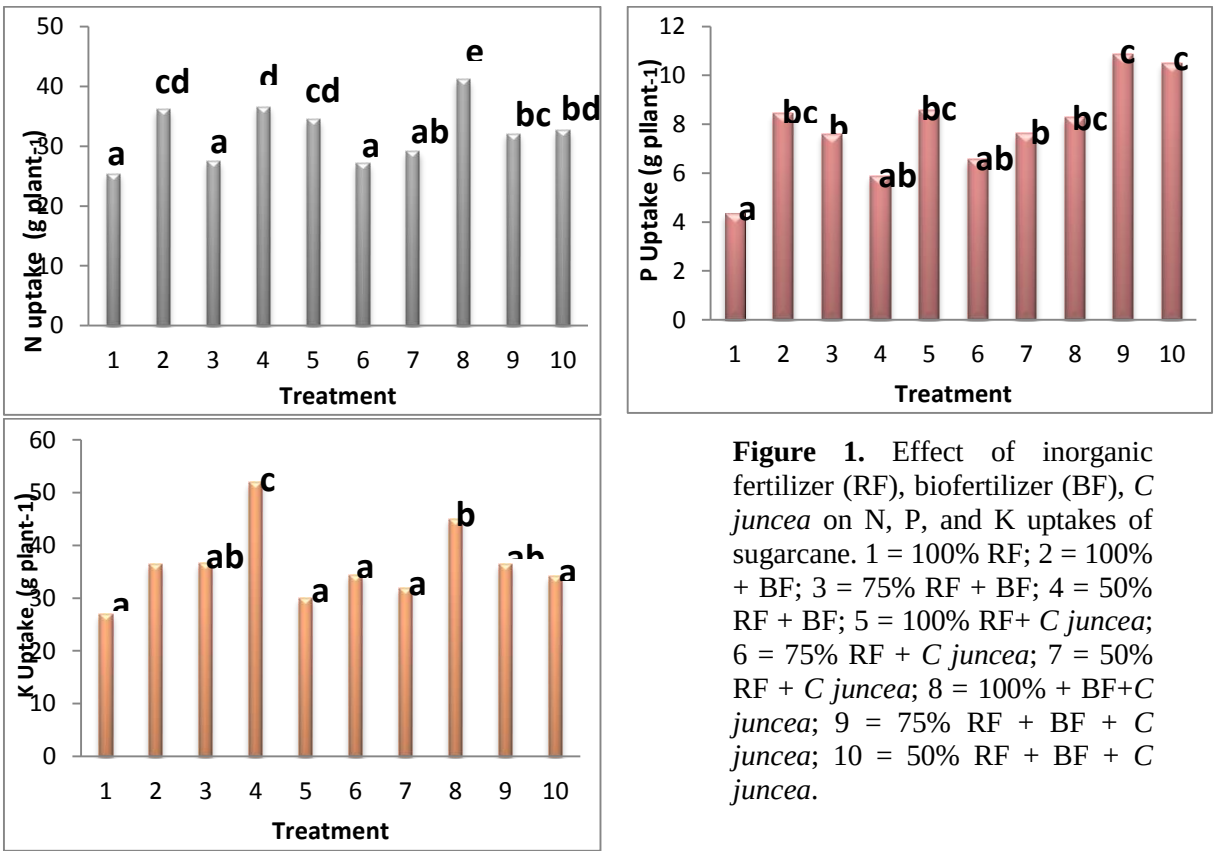

Figure 1. Effect of inorganic fertilizer (RF), biofertilizer (BF), *C juncea* on N, P, and K uptakes of sugarcane. 1 = 100% RF; 2 = 100% + BF; 3 = 75% RF + BF; 4 = 50% RF + BF; 5 = 100% RF+ *C juncea*; 6 = 75% RF + *C juncea*; 7 = 50% RF + *C juncea*; 8 = 100% + BF+*C juncea*; 9 = 75% RF + BF + *C juncea*; 10 = 50% RF + BF + *C juncea*.

For P uptake, treatment 9 (75% RF + BF + *C juncea*) caused sugarcane absorbed the highest amount of P (10.87 g plant^{-1}) and increased P absorption 1.40 times if compared to the control. However, treatment 10 (50 % RF + BF + *C juncea*), 2 (100% RF + BF), 5 (100% RF+ *C juncea*) and 8 (100% +

BF+*C juncea*) were not significantly different to the treatment 9. On parameter of K uptake, the highest K content in sugarcane plant (51.90 g plant^{-1}) was found in treatment 4 (50% RF + BF). Comparing to the control, treatment 4 (50% RF + BF + *C juncea*) increased K uptake by 92.94%, but not significantly different to the treatment 8 (100% RF + BF + *C juncea*).

3.2. Growth of Sugarcane

As expected, growth of sugarcane was affected by addition of biofertilizer and *C juncea* together with inorganic fertilizer at different rate. The effect of treatment on length of stalk and plant population of sugarcane was presented in Figure 2. Sugarcane which had the longest of stalk (343 cm) was found when it received 100% recommended inorganic fertilizer with biofertilizer and *C juncea* (treatment 8), but it was not significantly different with control. Addition of biofertilizer and *C juncea* with lower rate of inorganic fertilizer significantly decreased the length of stalk, the shortest of stalk length of sugarcane was found at the lowest rate of inorganic fertilizer which was added together with biofertilizer plus *C juncea* (treatment 10).

For the plant population parameter, the most dense stalk of sugarcane (17.55 stalk m^{-1}) was found at treatment 2 (100% RF + BF) indicating that addition of biofertilizer together with 100% recommended inorganic fertilizer produced the most amount of sugarcane stalk. The treatment increased the number of stalk by 12.06% over the control treatment. However treatment 10 (50% RF + BF + *C juncea*) was not significantly different to treatment 2.

The effect of treatments on diameter of stalk and number of internodes per stalk were presented in Figure 3. Addition of biofertilizer on the 100% RF with or without *C juncea* significantly decreased diameter of stalk. Therefore the biggest diameter of sugarcane stalk (25.41 mm) was observed at sugarcane with 100% recommended inorganic fertilizer (control treatment). However the diameter of stalk size was not significantly different to that with addition of biofertilizer with 75% recommended inorganic fertilizer (treatment 3). For the number of internodes per stalk, the highest number of internodes (27.31 internodes) was found at sugarcane plots where biofertilizer added and 75% were added (treatment 3). The number of internodes per stalk of the treatment was 0.64% higher than that of control treatment.

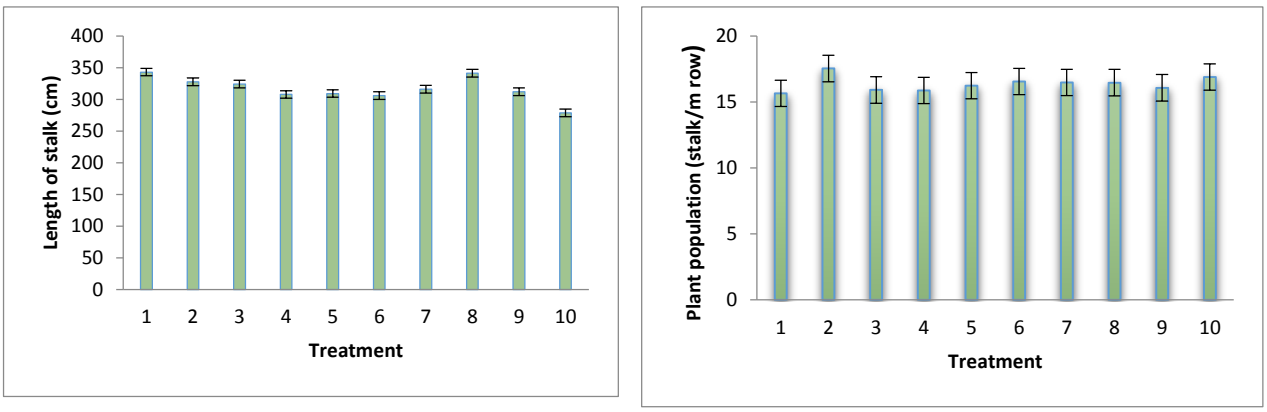

Figure 2. Effect of inorganic fertilizer (RF), biofertilizer (BF), *C juncea* (Cj) on length of stalk and plant population of sugarcane. 1 = 100% RF; 2 = 100% + BF; 3 = 75% RF + BF; 4 = 50% RF + BF; 5 = 100% RF+ *C juncea*; 6 = 75% RF + *C juncea*; 7 = 50% RF + *C juncea*; 8 = 100% + BF+ *C juncea*; 9 = 75% RF + BF + *C juncea*; 10 = 50% RF + BF + *C juncea*.

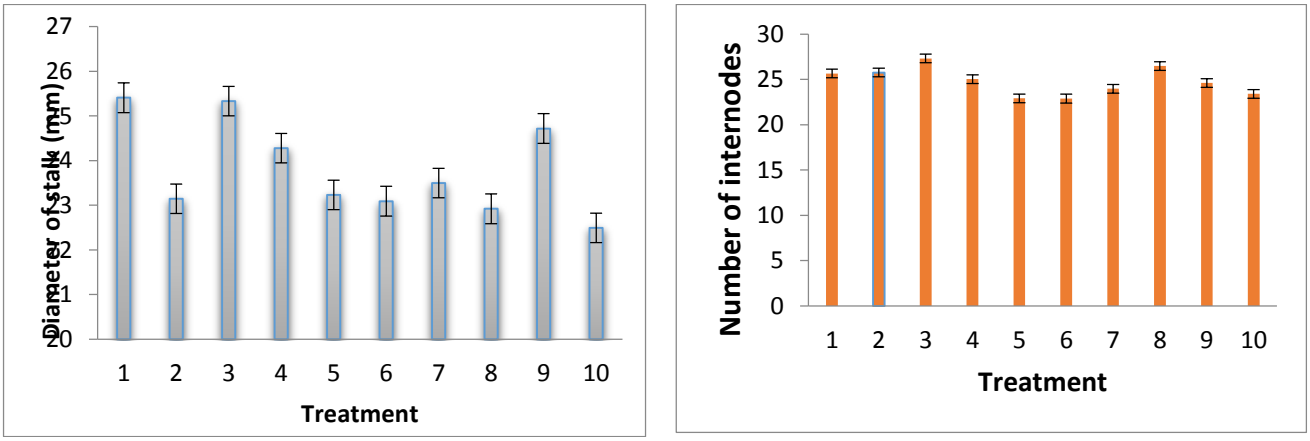

Figure 3. Effect of inorganic fertilizer (RF), biofertilizer (BF), *C juncea* (Cj) on diameter of stalk and number of internode of sugarcane. 1 = 100% RF; 2 = 100% + BF; 3 = 75% RF + BF; 4 = 50% RF + BF; 5 = 100% RF+ *C juncea*; 6 = 75% RF + *C juncea*; 7 = 50% RF + *C juncea*; 8 = 100% + BF+ *C juncea*; 9 = 75% RF + BF + *C juncea*; 10 = 50% RF + BF + *C juncea*.

4. Discussion

This research demonstrated that addition of biofertilizer with or without *C juncea* and inorganic fertilizer together raised nutrients uptakes and increased growth of sugarcane to some extent. Addition of biofertilizer with *C juncea* provided soil nutrients more available to the plant which was indicated by increasing N, P and K uptakes by sugarcane. In our research, addition of biofertilizer plus *C juncea* together with inorganic fertilizer at different rate caused the highest absorption of N, P and K nutrients. For a comparable research in which *Azotobacter* inoculant used as biofertilizer, an increase of N, P, and K uptake by *Zea mays* was associated with addition of biofertilizer and organic matter [13]. Similar finding was reported by previous researcher who observed that N, P, K nutrients uptake by *Zea mays* were raised by addition of biofertilizer (containing mix of *Azotobacter chroococcum* and phosphate solubilizing bacteria) together with *C juncea*.

Nitrogen (N), phosphorus (P) and potassium (K) are macro nutrients which have important role in supporting growth and yield of sugarcane. Nitrogen is an important nutrient in plant metabolism due its function in the synthesis of proteins and other organic compound so that it is essential for vigorous, vegetative growth and yield of sugarcane [5]. Phosphorus nutrient is required in energy metabolism, synthesis of nucleic acids, photosynthesis, respiration and enzyme regulation [16]. Thus, by adding biofertilizer of N fixing and P solubilizing bacteria together with inorganic fertilizer will raise growth of sugarcane.

The growth of sugarcane as it was expressed by length of stalk, plant population and number of internodes parameters was elevated by addition of biofertilizer with or without *C juncea*. Similar results of positive effect of biofertilizer on growth of sugarcane were reported by some workers [7; 15, 21]. Higher growth of sugarcane due to addition of biofertilizer with or with no *C juncea* than control (addition of inorganic fertilizer only) might be due to increasing N, P and K nutrition up take. In this current research, there was significant regression between N uptake and length of stalk (R^2 = 0.56), increasing in N uptake would increase the length of stalk (Figure 4). In the research, we found that combination of 100% recommended inorganic fertilizer with biofertilizer and *C juncea* gave the highest of N and length of stalk.

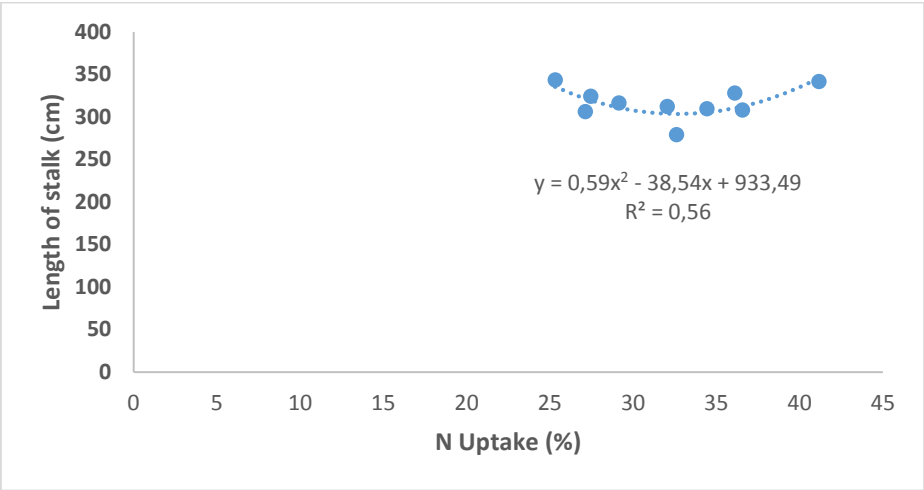

Figure 4. Regression between N uptake and length of stalk of sugarcane

5. Conclusion

Addition of biofertilizer containing N fixer and P solubilizer bacteria and *C juncea* together with recommended fertilizer raised N, P, and K uptakes and growth of sugarcane. The combination of biofertilizer and *C. juncea* with 100% rate of recommended fertilizer of NPK gave the highest of N uptake and length of stalk of sugarcane.

References

[1] Bhuiyan M K I., Riko C M, Mintah L O, Kim M K, Son T K, Chung I K and Lee S C 2006 Effect of biofertilizer on growth and yield of rice *Korean J. Crop Sci.* **51** 282–6.

[2] Bokhtiar, S, Gafur M and Rachman A 2003 Effect of *Crotalaria* and *Sesbania aculeate* green and N fertilizer on soil fertility and productivity of sugarcane *The Journal of Agricultural Science* **140** 3205-9

[3] Cantarella H and Rossetto R 2010 Fertilizers for Sugarcane *Sugarcane Bioethanol*, ed L A B Cortez (Sao Paulo: Blucher) p 954

[4] Castro F O, Contreras M G L, Cauich M T, Cabrera O A G, Mendez J M, Franco A D, Sánchez R, and Rocha R J 2013 Response of rice cultivation to biofertilizers in Campeche, Mexico *Agricultural Sciences* **4** 715-20

[5] Costa A R F C, Rolim M M, Bonfim-Silva E M, Neto D E S, Pedrosa E R M and Silva E F F 2016 Accumulation of nitrogen, phosphorus and potassium in sugarcane cultivated under different types of water management and doses of nitrogen *Australian Journal of Crop Science* **10** 362–9.

[6] Eviati and Sulaiman 2012 *Analysis of Soil Chemical, Plant, Watere and Fertilizer* (Bogor: Indonesian Agency for Agriculture Research and Development)

[7] Gosal S K, Kalia A, Uppal S K, Kumar R, Walia S S, Singh K and Singh H 2012 Assessing the benefits of azotobacter bacterization in sugarcane: A Field Appraisal *Sugar Tech* **14** 61–7

[8] Iwuagwu M, Chucwuka K, Uka U N, and Amandianeze M C 2013 Effects of biofertilizers on the growth of *Zea mays* L *Asian Journal of Microbiology, Biotechnology and Environmental Sciences* **15** 235-40.

[9] Javaid A 2011 Effects of biofertilizers combined with different soil amendments on potted rice plants *Chilean Journal of Agricultural Research* **71** 157-63.

[10] Kabiraj R C, Islam M S, Siidiqui M N and Siddika A 2017 Effect of intercrop green manuring and nitrogen on the growth, yield and quality of sugarcane (*Saccharum officinerum* L.) *J. Agrofor. Environ* **11** 257–60.

[11] Kalhapure A H, Shete B T and Dhonde M B 2013 Integrated Nutrient Management in Maize (*Zea Mays* L.) for Increasing Production with Sustainability *International Journal of Agriculture and Food Science Technology* **4** 195-206

[12] Marngar E and Dawson J 2017 Effect of Biofertilizers, Levels of Nitrogen and Zinc on Growth and Yield of Hybrid Maize (*Zea mays L.*) *International Journal of Current Microbiology and Applied Sciences* **6** 3614-362.

[13] Meena D, Tiwari D, Chaudhari S K, Biswas D R, Narjary B, Meena A L, Meena B L and Meena R B 2013 Effect of Biofertilizer and Nutrient Levels on Yield and Nutrient Uptake byMaize (*Zea mays* L.) *Annals of Agri-Bio Research* **18** 176-81

[14] Oliveira R I, Medeiros M R F A, Freire C S, Freire F J, Neto D E S and Oliveira E C A 2016 Nutrient partitioning and nutritional requirement in sugarcane *Australian Journal of Crop Science* **10** 69-75.

[15] Oliveira F L N, Oliveira W S, Stamford N P, Silva E V N, Santos C E R S and Freitas A D S 2017 Effectiveness of biofertilizer enriched in N by *Beijerinckia indica* on sugarcane grown on an Ultisol and the interactive effects between biofertilizer and sugarcane filter cake *Journal of Soil Science and Plant Nutrient* **17** 1030-57.

[16] ``Raghothama K G 1999 Phosphate acquisition *Ann Rev Plant Physiol Plant Mol Bio* **50** 665-93.

[17] `Roba T B 2018 Review on: The Effect of Mixing Organic and Inorganic Fertilizer on Productivity and Soil Fertility *Open Access Library Journal*, 5: e4618.

[18] Santana, A A D de; Moura R M de and Pedrosa E M R 2003 Effect of crop rotation with sugarcaneand *C. juncea* on plant parasitic nematodes of yam *Nematologia Brasileira* **27** 13-6.

[19] Shen H, He X, Liu Y, Chen Y, Tang J and Guo T 2016 A Complex Inoculant of N_2-Fixing, P- and K-Solubilizing Bacteria from a Purple Soil Improves the Growth of Kiwifruit (*Actinidia chinensis*) Plantlets *Frontiers in Microbiology* **7** 841

[20] Stamford N P, Junior S, Santos C E R S, Freitas A D S, Santos C M A, Arnaud T M S and Soares H R 2014. Yield of grape (*Vitis labrusca*cv. Isabel) and soil nutrients availability affected by biofertilizers with *diazotrophic* bacteria and fungi chitosan *Austr. J. Crop Sci.* **8** 301-6.

[21] Stamford N P, Neto D U S, Freitas A D S, Oliveira E C A, Oliveira W S and Cruz L 2015 Rock biofertilizer and earthworm compost on sugarcane performance and soil attributes in two consecutive years *Scientia Agricola* **73** 29 – 33.